MILITARY STANDARD

MIL-STD-1325-188
(NAVY)

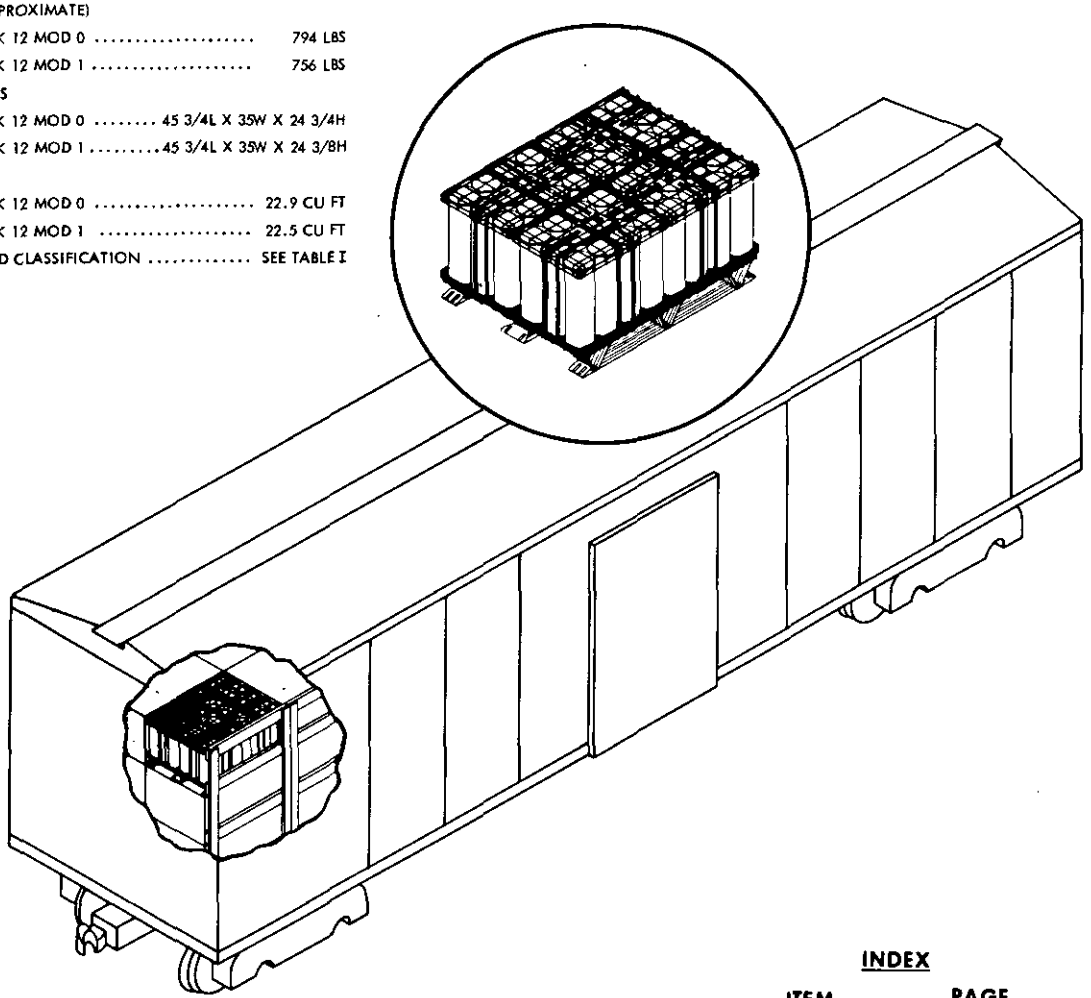
24 JUNE 1977

RAILCAR LOADING OF HAZARDOUS MATERIALS

CARTRIDGE, 4.4 INCH, CHAFF, MK 171 TYPE, MK 173, MK 178, MK 179 (RBOC)
IN PALLET ADAPTER MK 132 MOD 0
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/296
WEIGHT (APPROXIMATE)
PALLET MK 12 MOD 0 ..... 794 LBS
PALLET MK 12 MOD 1 ..... 756 LBS
DIMENSIONS
PALLET MK 12 MOD 0 ..... 45 3/4L X 35W X 24 3/4H
PALLET MK 12 MOD 1 .....45 3/4L X 35W X 24 3/8H
CUBE
PALLET MK 12 MOD 0 ..... 22.9 CU FT
PALLET MK 12 MOD 1 ..... 22.5 CU FT
DOT HAZARD CLASSIFICATION ..... SEE TABLE I



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCK-LOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

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FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M. S. Gray NWHC 6/16/77
TECHNICAL DIRECTION AGENT (TDA) DATE

SEA SYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grassmuck 5/26/77
SIGNATURE SUPERVISOR, MILITARY AND INTERMODEL SERVICES DATE

ORIGINATOR S R Petrie 6/15/77 DATE
NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL NOT BE USED FOR ITEMS HAVING EXPLOSIVE A OR B HAZARD CLASSIFICATION.
3. LOADING PLANS SHOWN ARE FOR COMMERCIAL BOXCARS WITH 40 FT. 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT. 6 INCHES LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF CARTRIDGE, 4.4 INCH, CHAFF, MK171 TYPE, MK 173, MK 178, MK 179 (RBOC) IN PALLET ADAPTER MK 132 MOD O PALLETIZED IN ACCORDANCE WITH WR-54/296.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:

    DUNNAGE LUMBER-FED. SPEC MM-L-751

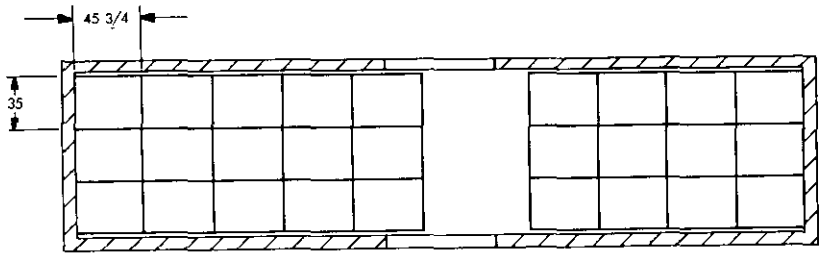
    NAILS-FED. SPEC FF-N-105

    STRAPPING-FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).

    SEALS-FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA. CLOSE AND SEAL BOXCAR DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 VOL. I.

40 FT 6 IN. BOXCAR, COMMERCIAL

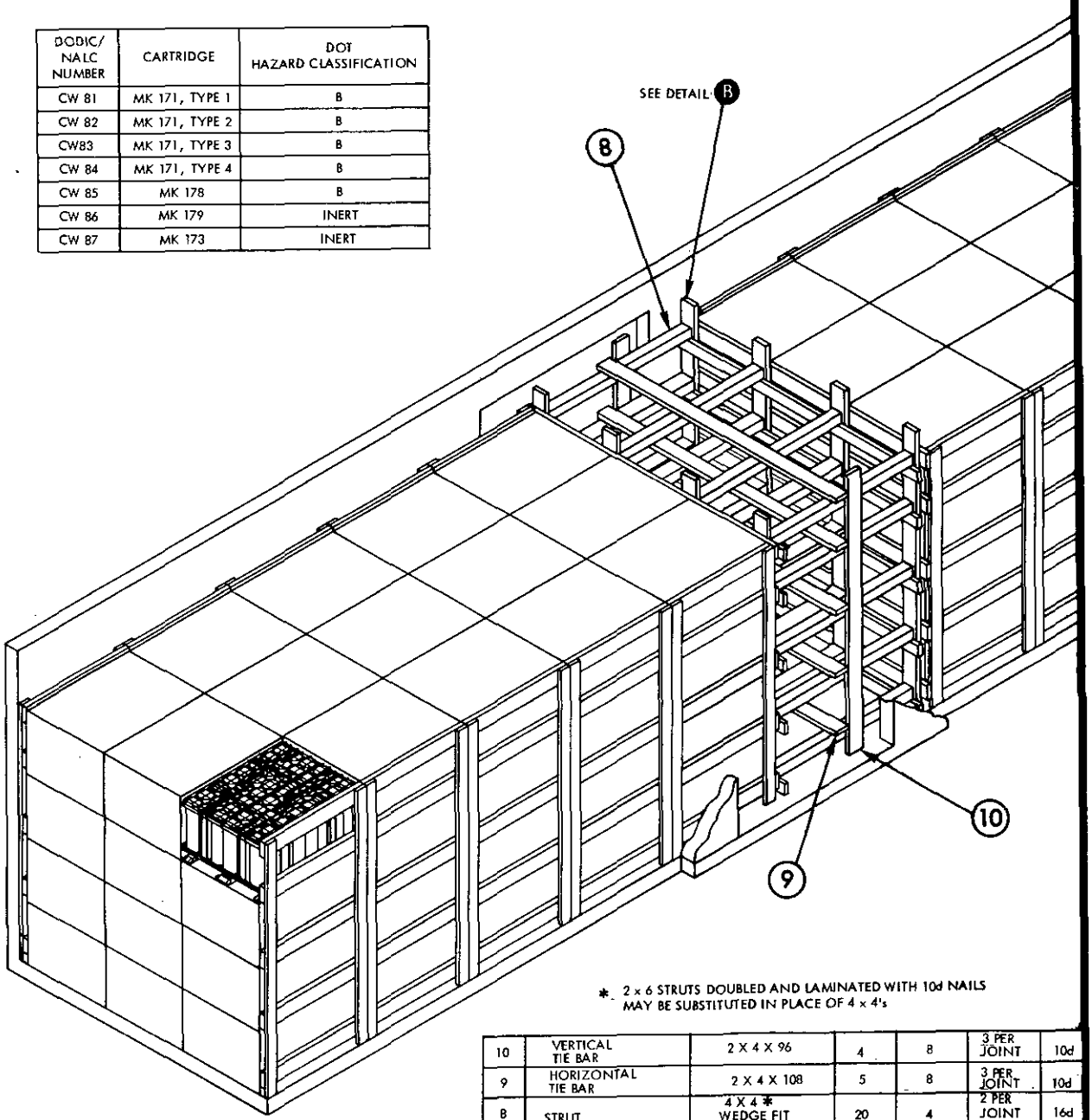
1. THE CARLOAD CONSISTS OF 108 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT. WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT. WIDE.
4. WHEN BOXCARS WIDER THAN 9 FT. 2 IN. ARE USED, INCREASE THE THICKNESS OF PIECE 1 BY LAMINATING AN APPROPRIATELY THICK PIECE TO IT. IF THE WIDTH OF BOXCAR IS LESS THAN 9 FT. 2 IN. DECREASE THE THICKNESS OF PIECE 1 AS NECESSARY.



LOADING PLAN

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DODIC/ NALC NUMBER	CARTRIDGE	DOT HAZARD CLASSIFICATION
CW 81	MK 171, TYPE 1	B
CW 82	MK 171, TYPE 2	B
CW83	MK 171, TYPE 3	B
CW 84	MK 171, TYPE 4	B
CW 85	MK 178	B
CW 86	MK 179	INERT
CW 87	MK 173	INERT



* 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4's

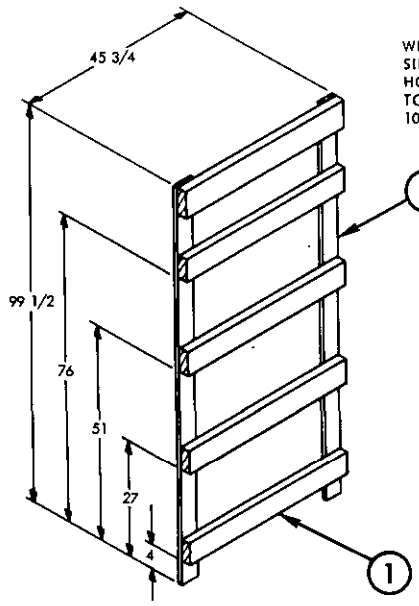
CARLOAD DATA

	PALLET MK 12-0	PALLET MK 12-1
NO. OF UNIT LOADS	108	108
LOAD WEIGHT (APPROXIMATE)	85752 LBS.	81648 LBS.
DUNNAGE WEIGHT (APPROXIMATE)	2090 LBS.	2090 LBS.
CARLOAD WEIGHT (APPROXIMATE)	87842 LBS.	83738 LBS.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	VERTICAL TIE BAR	2 X 4 X 96	4	8	3 PER JOINT	10d
9	HORIZONTAL TIE BAR	2 X 4 X 108	5	8	3 PER JOINT	10d
8	STRUT	4 X 4 * WEDGE FIT	20	4	2 PER JOINT	16d
7	HOLDDOWN	2 X 3 X CUT TO SUIT	4	6	3	10d
6	HOLDDOWN SPACER	2 X 4 X CAR WIDTH-1	2	4	3 PER JOINT	10d
5	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-1	10	4	3 PER JOINT	10d
4	CENTER GATE VERTICAL	2 X 6 X 108	8	SEE 3	–	–
3	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	10	4	3 PER JOINT	10d
2	SWAY BRACE VERTICAL	1 X 4 X 99 1/2	36	1	2 PER JOINT	6d
1	SWAY BRACE HORIZONTAL	2 X 6 X 45 3/4	90	SEE 2	–	–

LIST OF MATERIALS & NAILING DATA

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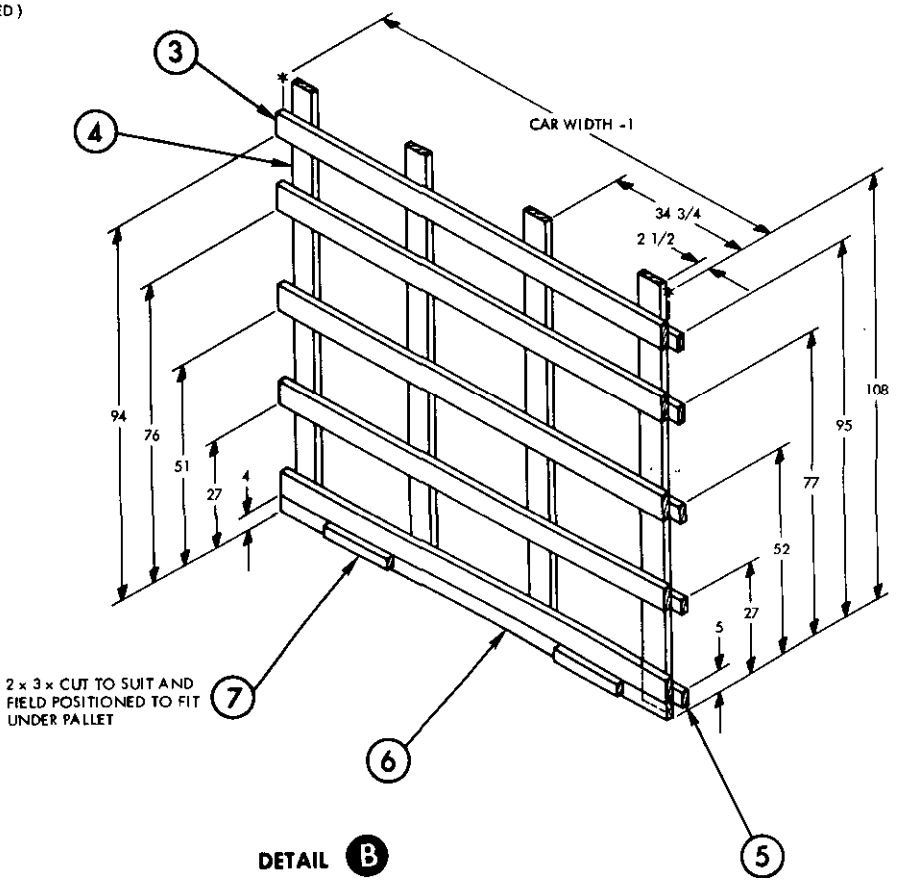
WHEN BOXCAR HAS WOOD SIDEWALLS, SWAY BRACE HORIZONTALS MAY BE NAILED TO THE SIDEWALL WITH ONE 10d NAIL PER FOOT.

PLACE VERTICAL AGAINST WALL

DETAIL A

SWAY BRACE ASSEMBLY

(18 REQUIRED)



DETAIL B

CENTER GATE

(2 REQUIRED)

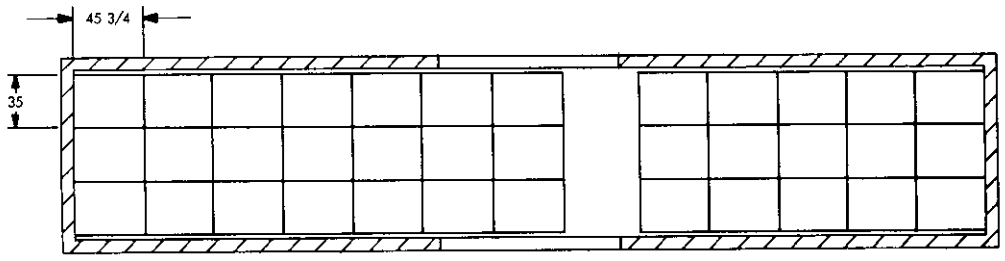
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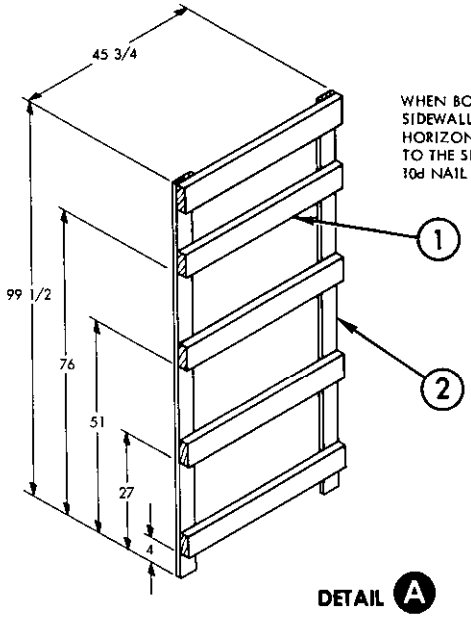
50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 144 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH THE "VERTICAL MEMBER AND STRAPPING" PROCEDURE OR THE "KNEE BRACING" PROCEDURE SHOWN IN MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT. WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT. WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS PROVIDING STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA ARE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. DIMENSIONAL LUMBER DOORWAY PROTECTION, PIECES 11 AND 12 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DIMENSIONAL LUMBER DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. PLUG DOORS MUST BE SECURELY CLOSED AND WIRED TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. WHEN BOXCARS WIDER THAN 9FT. 2 IN. ARE USED, INCREASE THE THICKNESS OF PIECE 1 BY LAMINATING AN APPROPRIATELY THICK PIECE TO IT. IF THE WIDTH OF BOXCARS IS LESS THAN 9FT. 2 IN. DECREASE THE THICKNESS OF PIECE 1 AS NECESSARY.



LOADING PLAN

LOAD LIMIT OF THE BOXCAR MUST BE 120,000 LBS. OR GREATER



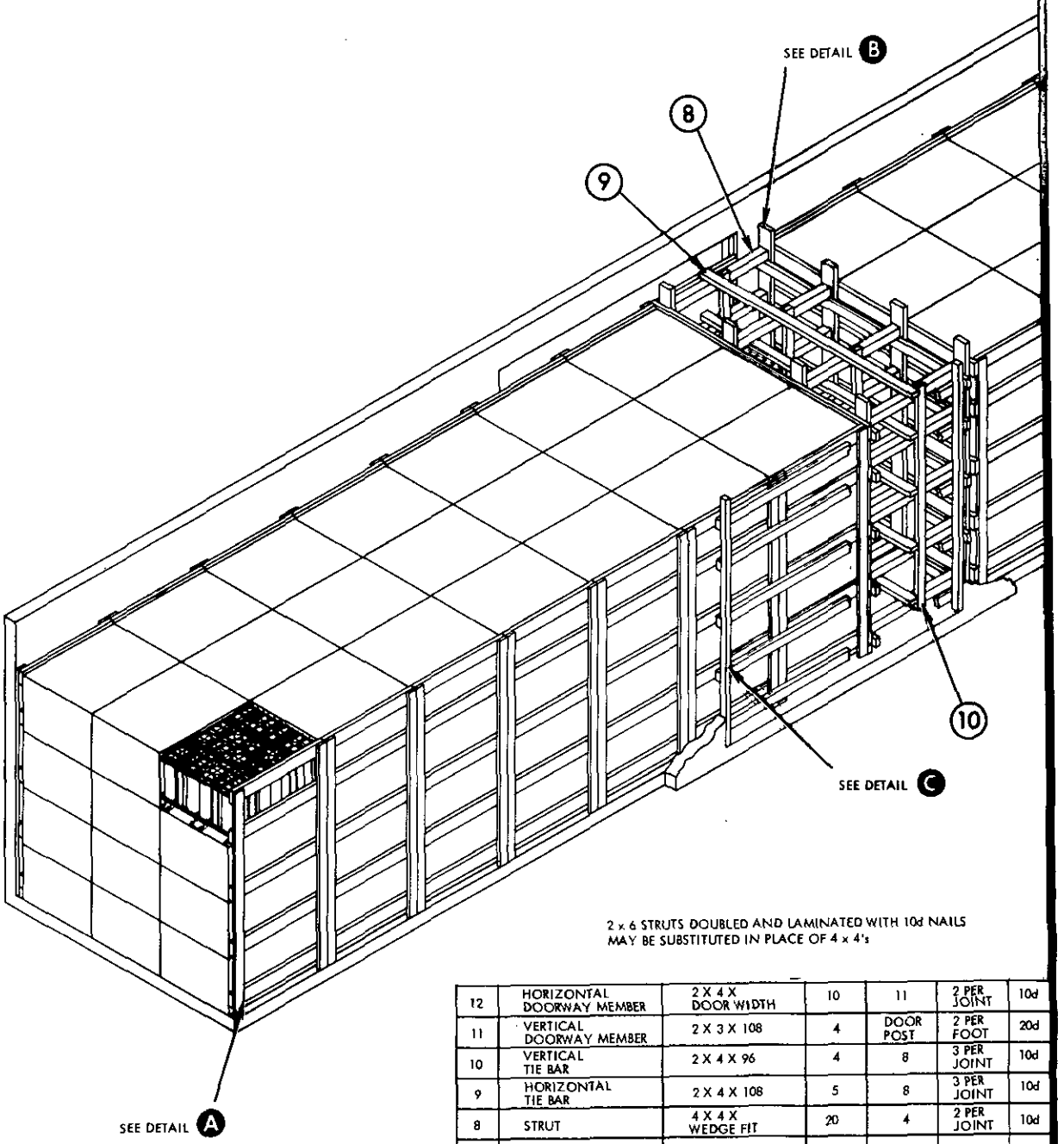
WHEN BOXCAR HAS WOOD SIDEWALLS, SWAY BRACE HORIZONTALS MAY BE NAILED TO THE SIDEWALL WITH ONE 10d NAIL PER FOOT.

PLACE VERTICAL AGAINST WALL

DETAIL A

SWAY BRACE ASSEMBLY

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2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4's

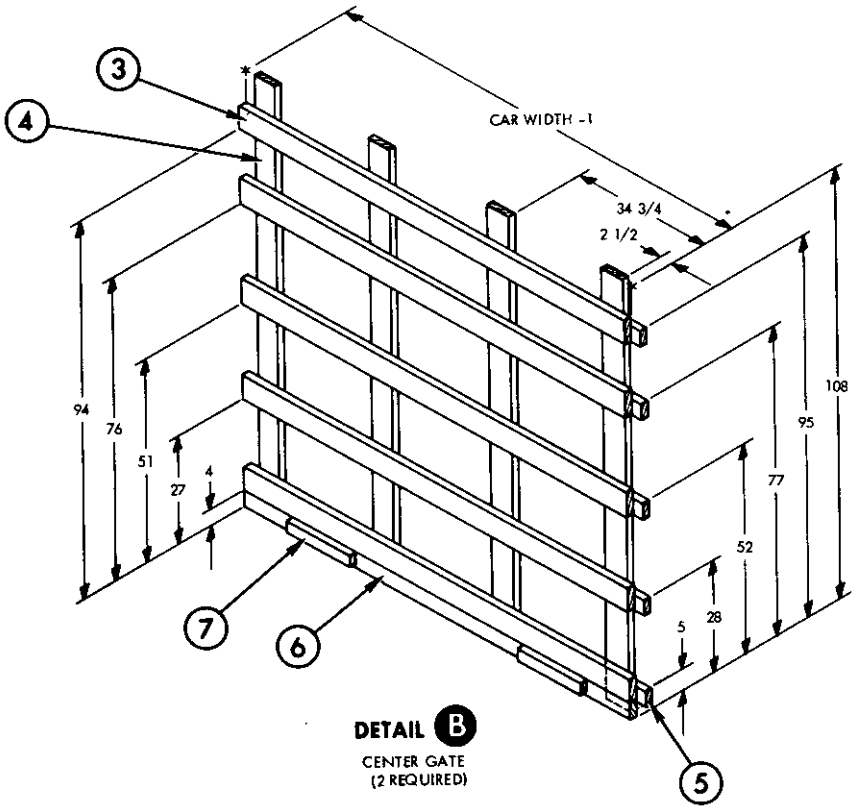
CARLOAD DATA

	PALLET MK 12-0	PALLET MK 12-1
NO. OF UNIT LOADS	144	144
LOAD WEIGHT (APPROXIMATE)	114336 LBS	108864 LBS
DUNNAGE WEIGHT (APPROXIMATE)	2412 LBS	2412 LBS
CARLOAD WEIGHT (APPROXIMATE)	116748 LBS	111276 LBS

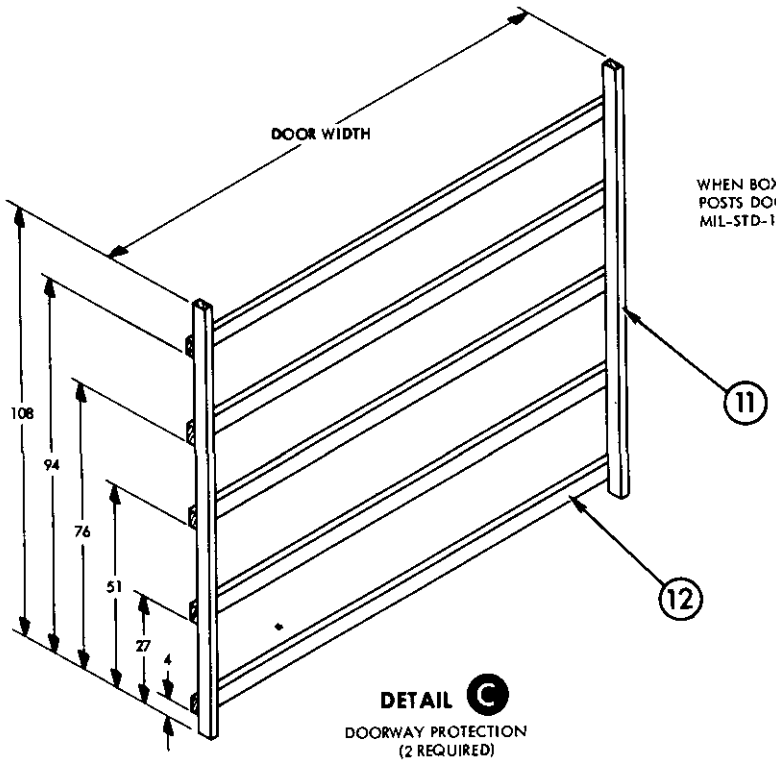
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
12	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	10	11	2 PER JOINT	10d
11	VERTICAL DOORWAY MEMBER	2 X 3 X 108	4	DOOR POST	2 PER FOOT	20d
10	VERTICAL TIE BAR	2 X 4 X 96	4	8	3 PER JOINT	10d
9	HORIZONTAL TIE BAR	2 X 4 X 108	5	8	3 PER JOINT	10d
8	STRUT	4 X 4 X WEDGE FIT	20	4	2 PER JOINT	10d
7	HOLD DOWN	2 X 3 X 18	4	6	3	10d
6	HOLDDOWN SPACER	2 X 4 X CAR WIDTH-1	2	4	3 PER JOINT	10d
5	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH-1	10	4	3 PER JOINT	10d
4	CENTER GATE VERTICAL	2 X 6 X 108	8	SEE 3	—	—
3	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH-1	10	4	3 PER JOINT	10d
2	SWAY BRACE VERTICAL	1 X 4 X 99 1/2	48	1	2 PER FOOT	6d
1	SWAY BRACE HORIZONTAL	2 X 6 X 45 3/4	120	SEE 2	—	—

LIST OF MATERIALS & NAILING DATA

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DETAIL B
CENTER GATE
(2 REQUIRED)



WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) IS REQUIRED.

DETAIL C
DOORWAY PROTECTION
(2 REQUIRED)

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO. 8340-N173)

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